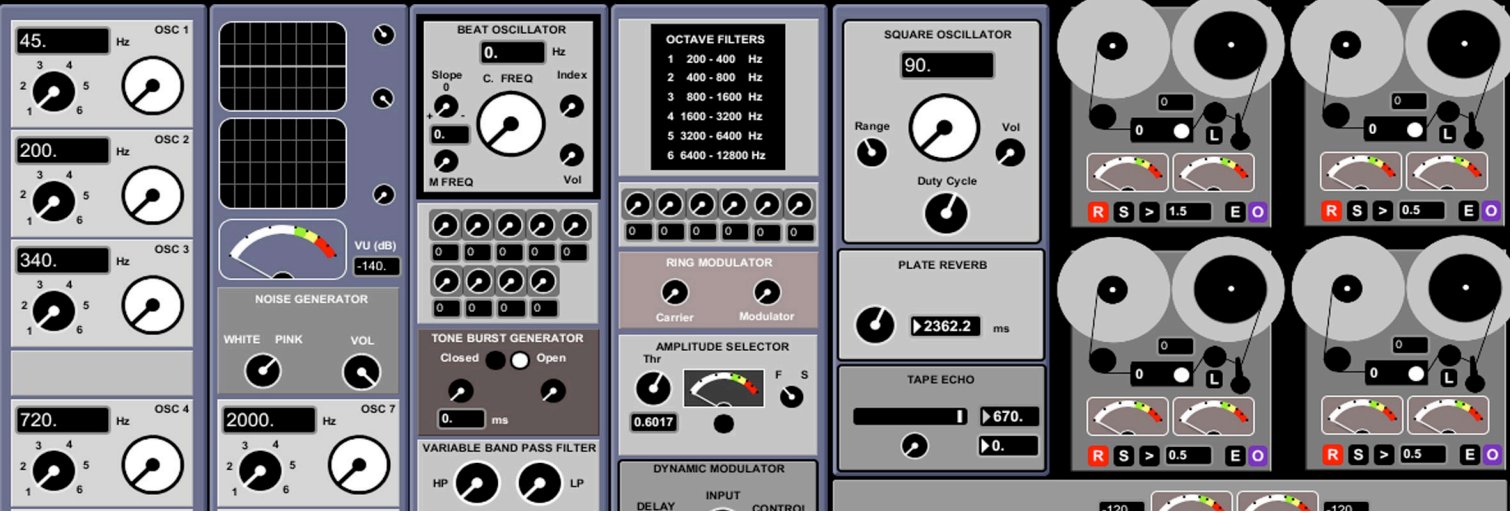
Berna

vintage electronic studio



PREFACE

Between the 1950s and the mid 1960s, long before Robert Moog and Wendy Carlos injected electronics into pop-music (with a few exceptions like the Barrons and Raymond Scott), electroacoustic music was pioneered by European radio laboratories and US universities. Composing with tapes and electronics was a serious painstaking and expensive affair, prerogative of a restricted elite of contemporary music composers and adventurous sound engineers. At that time there wasn't any electronic musical instruments market, as a matter of fact, most of the equipment was adapted from scientific tools belonging to radio engineering departments. Sometimes the equipment was built from scratch cannibalizing anything that had wires, tubes, and pots, more rarely, the studios used the few commercial instruments available in those days, such as the Melchord, the Trautonium and the Theremin. Contrarily to what happens today, electronic music in those days was everything but fast and easy to create. A few minutes of electronic composition could take more than one year of work. Everything was handmade, from complex timbres with multiple sine oscillator bounces (no keyboard was used), to tape editing with scissors and scotch-tape. Even sound envelopes were manually built by cutting tapes' edges at different

degrees of inclination. Ussachevsky's ADSR was yet to be invented!
As it has been stated many times by musicologists, engineers and composers, technologies had always a fundamental role in electronic music; it is difficult to say wheter technology had shaped music aestethics or vice versa. I believe both things are true. Let's take for example Moog synthesizers and keyboards. Before releasing his first commercial synthesizer, Robert Moog asked Vladimir Ussachevsky his opinion on placing a piano-keyboard into the design. Ussachevsky didn't agree. He thought that the keyboard would have limited the use of the synthesizer. Nevertheless Moog choosed to follow his instincts. But what would have happened if he didn't? Certainly Wendy Carlos would have never recorded Switched on Bach (at least with a Moog synthesizer) and this could have changed the whole history of modern music. On the other side, what if Stockhausen never thought of multichannel performances? Aestethics and technology are linked together in modern music much more than we realize, whether we listen to Daft Punk or Henry Pousseur.

Another interesting point lies in the limitations of technology. From the very beginning (let's say for example Edgar Varése's theories), composers and musicians had always dreamed to push the limits of technology further and further away in order to obtain a total

freedom of expression. This is the same dream that every software and hardware company has pursued (and sold) from the advent of the electronic musical instruments market. But what about intentionally placing some limits? When I didn't include SAVE and LOAD functions in Gleetchlab, a lot of people asked me why, for God's sake, I was making such an incomprehensible choice in the digital era. The reason was pretty simple and somehow revolutionary: by erasing a granted technology (to save and load pacthes) I've placed people in a (paradoxically) groundbreaking situation. Basically I've forced them to start from scratch every time they run the software, just like any analog equipment. Many people were grateful and feel inspired by that. Of course such possibility was there everytime in any software. They could have avoided to save their work while using their favourite softwares, but they didn't, just because that technology was right there. In other words, ideas do not necessarily come from the absence of limits. Sometimes limits push you to think in a different way. This came to my mind again recently while listening to Bruno Maderna's electronic music. At a certain point I've asked myself what would happen if modern musicians could use a late fifties electronic music laboratory? Beside the technological gap there is a substantial difference between that and a bunch of early seventies commercial synthesizers (like Moog, Arp, Ems, Korg, and so on) and that is aesthetics. Early electronic music composers weren't interested in a

keyboard (which leads to traditional) approach, they wanted to compose music by sculpting the sound from within. (and they had very few instruments to do so). They wanted to articulate masses of sound, understand sound, get rid of fixed intervals, dig and manipulate sound in its smallest elements with scientific precision. That is why they preferred to manually tune oscillators instead of playing them on a keyboard. No need to say that such equipment looks prehistoric compared to a Minimoog. Sound laboratories of the late fifties lacked many of the well known devices you can find in any synthesizer produced from 1971 to nowadays. No envelopes, no control voltage, no modulation busses, not to talk of very basic waveforms and modulations. What is the point then, in programming a simulation (more or less accurate, because any digital process is a numerical approximation of reality) of such archaic musical equipment? The answer is twofold. The first is that I'm curious to hear what today's musicians could do with it. I do not expect people to make serial music in 2009 (however it would be interesting to hear some), instead I think that many could came up with fresh new ideas, stimulated by the radical differences between Berna and the modern electronic music tools. This is a sort of experiment and I encourage every Berna user to send me music files of their work. Another good reason is that this software may be useful to academy students that want to take a closer look to serial and tape music wit-

hout having to learn complex softwares like Max/MSP, PD or C-Sound. Of course many of the possibilities of Berna can be found elsewhere in many softwares, however, this application contains everything[1] you need to produce and study such music syntax and the layout is a good approximation of what you could have found in the Cologne's WDR or Milano's RAI Phonology Studio.

As a final note to this preface I need to clearly state that this software is not intended to emulate a vintage sound. You will not hear a simulation of digitally aged oscillators, filters and tape compression. This is a serious learning and creative tool for musicians who want to explore the way of the early days of electronic music.

Giorgio Sancristoforo
Milan, October 21st, 2009

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[1] Except tape editing. For that you will need a DAW like Cubase, Nuendo, Logic, Protools, Soundtrack and the like...

IN DEMO MODE THE SOFTWARE WILL RUN FOR 8 MINUTES

HOW TO AUTHORIZE YOUR SOFTWARE

1) Download Berna from www.gleetchplug.com
2) Pay for the software with Paypal.
in 24/48 hours a serial number will be
sent to your email account (the one you have used to
register with Paypal)
3) In the main window click Authorization, a pop-up
window will open:
a) Input your Paypal email address
b) Input the serial number (with spaces)

For problems regarding the authorization
process, please write to
sales@gleetchplug.com

Please try the software before you buy.

CHAPTER 1
Setting the soundcard:

Before using Berna for the first time, take a few moments to set the soundcard options. Click “AUDIO OPTIONS” in the main window and check the following:

Audio: Should be ON (Usually this should be switched ON automatically at startup).

Driver: Select your soundcard to determine audio input and output.

Sampling rate: Sets the desired sampling rate.

“The I/O Vector Size (I/O stands for input/output) controls the number of samples that are transferred to and from the audio interface at one time. The Signal Vector Size sets the number of samples that are calculated by MSP objects at one time. This can be less than or equal to the I/O Vector Size, but not more. The I/O Vector Size may have an effect on latency and overall performance. A smaller vector size may reduce the inherent delay between audio input and audio output, because MSP has to perform calculations for a smaller chunk of time. On the other hand, there is an additional computational burden each time MSP prepares to calculate another vector (the next chunk of audio), so it is easier overall for the

processor to compute a larger vector. However, there is another side to this story. When MSP calculates a vector of audio, it does so in what is known as an interrupt. If MSP is running on your computer, whatever you happen to be doing (word processing, for example) is interrupted and an I/O vector's worth of audio is calculated and played. Then the computer returns to its normally scheduled program. If the vector size is large enough, the computer may get a bit behind and the audio output may start to click because the processing took longer than the computer expected. Reducing the I/O Vector Size may solve this problem, or it may not. On the other hand, if you try to generate too many interrupts, the computer will slow down trying to process them. Therefore, you'll typically find the smaller I/O Vector Sizes consume a greater percentage of the computer's resources. Optimizing the performance of any particular signal network when you are close to the limit of your CPU's capability is a trial-and-error process. That's why MSP provides you with a choice of vector sizes."
From the max5 Manual

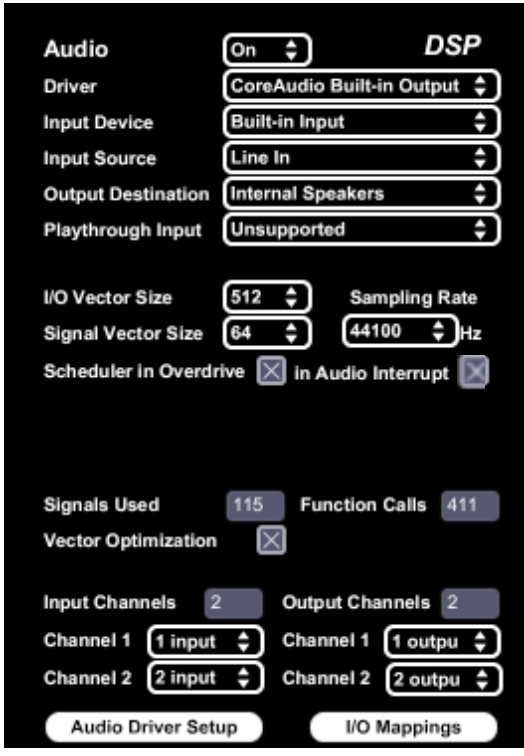
Max Scheduler in Overdrive and Scheduler Audio Interrupt should both be ON.

Input /Output Channels: Select the desired stereo IN/OUT couples. Other Audio IN/OUT options for multi I/O soundcards are located in the I/O Mappings menu.

Vector Optimization: Should be ON

CPU Limit: Should be 0%

Audio driver Setup button: your soundcard setup page.



CHAPTER 2 OVERVIEW

The Berna user interface is very simple. It has been designed with the intent of emulating the layout of an old electronic music studio. There are 5 different kind of tools in it:

1) Generators
2) Sound processors
3) Signal monitoring tools
4) Tape recorders
5) Mixer

On the top right you can find also:

1) An ON/OFF switch: turns Berna's Audio ON or OFF
2) Matrix: the patchbay of the studio
3) Audio Options: a window where you can set your sound card preferences
4) Authorization: to unlock the DEMO MODE.
5) The pattern grid
6) Load pattern button
7) Save pattern button

The pattern grid is a simple straightforward way to store and recall 30 patterns (and you can save them all on your Mac).

Their use is very simple: To store a pattern shift+mouse click a dot. To recall a pattern mouse click a dot. White dots are clean patterns. Red dots are stored patterns. A blue dot is the current recalled pattern.
Stored patterns can be overwriten.

You can store any setting of Berna, except the files that have been loaded into the tape players. You will have to load them manually from the tape players.

Record & Play functions on tape players are not stored.

To save all your presets on your hard drive press SAVE, a save dialog will open.
To load them... well, press LOAD





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CHAPTER 3 MATRIX

The Matrix is the most important feature to understand how Berna works. The Matrix makes Berna modular software. Instead of using graphical patch cords, a matrix is a simple and clean way to route all processor ins and outs. Remember: OUTPUTS are on TOP of the grid, INPUTS are on the left of the grid.

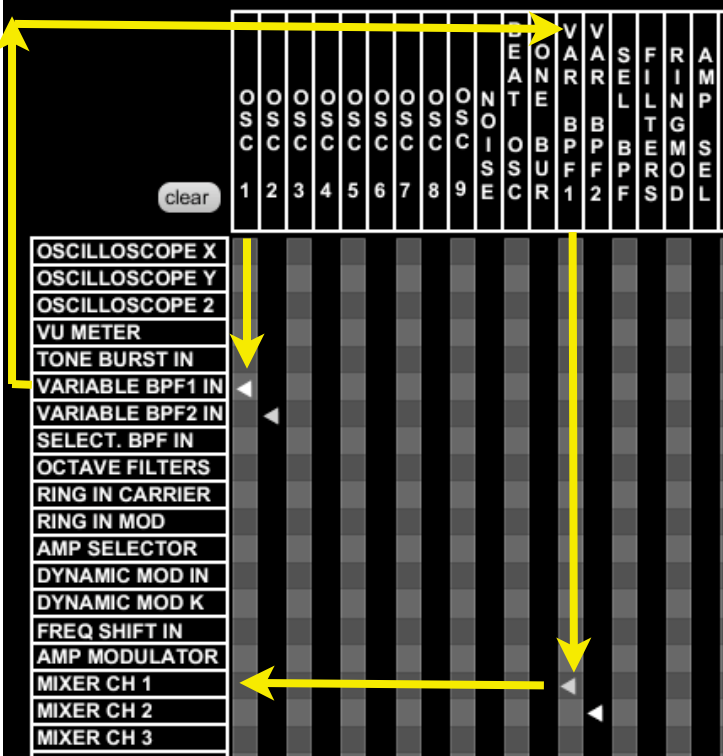
WITHOUT PATCHING THE MATRIX YOU CAN'T HEAR OR RECORD ANYTHING!

To hear audio coming from the oscillators or processors, you must patch signals to one of the mixer channells. Only the 4 tape recorders can play without having to be patched (although it is possible), from the mixer routing options.

All the signals in the matrix are MONO.
The only stereo signals in Berna are the Mixer's MASTER FADER and the 4 tape recorders output (when not patched to the matrix) in the MIXER.

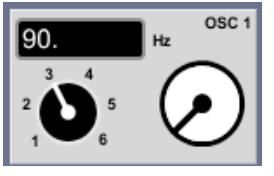
In the example here, two oscillators are patched each one to two variable band pass filters. The filters' output then is patched to mixer's channel 1 and 2. The yellow arrows show clearly the routing path of Oscillator 1

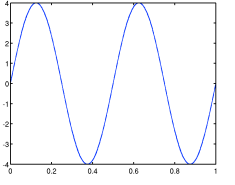
If you want to patch external signals form you soundcard use EXT IN 1 and EXT IN 2



CHAPTER 4 GENERATORS

During the 1950s the most common technique used to create and shape electronic timbres was additive synthesis. Complex waveforms were synthesized adding sinuoidal waveforms with different frequencyes and amplitude. In Berna you have 9 sine oscillators. All of them are identical, however I wanted to respect some specifications. The left dial is a 6 positions switch that sets the range of the right dial (frequency). Each of the nine sinusoidal oscillators has different ranges. This way, the oscillators are "tuned" in a way similar to acoustic musical instruments. For the ranges of the oscillators I've followed the original ranges of the RAI Phonology Studio's oscillators created by Alfredo Lietti[2] around 1956.



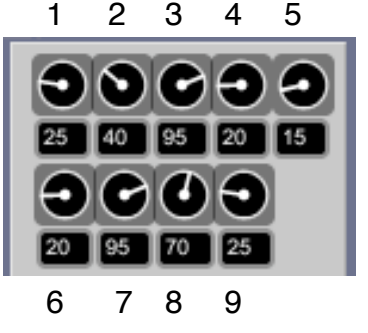


[2] Lo studio di Fonologia, un diario musicale 1954-1983 - Ricordi Publishing Milano 2009 - page 53

While these settings are important historical information, I've decided to include a very precise frequency display (to the 5th decimal). This is not just a display, frequencies can be set in the oscillator by clicking and dragging the mouse on it (if you point the mouse on a decimal number you will obtain a very fine tuning) or with the computer's keyboard. With this method all the 9 oscillators can output any frequency between 0 and 20000 Hz.

The nine oscillators are hardwired to a small mixer placed in the layout. This device controls the output volume of each sinusoidal oscillator, numbered from left to right, from top to bottom.

USE THIS IF YOU WANT TO HEAR THE 9 OSCILLATORS!



The Berna's noise generator outputs two different kind of noises: white noise and pink noise.

White noise signal has equal power in any band of a given bandwidth. For example the band between 50 and 100 Hz has the same energy of the band between 500 and 550Hz.

Pink noise signal has equal power in bands that are proportionally wide. For example the band between 50 and 100Hz has the same energy of the band between 1000 and 2000Hz. We could say that pink noise is white noise filtered with a 3dB/Oct low pass filter.

Noise can be filtered and refined to sound like such things as the wind and the ocean, and is a rich source of background sound and texture for the composer of electronic music[3]



[3] Electronic and Experimental Music by Tom Holmes - Routledge Publishing - 1985 - pag 17

The beat oscillator is a sinusoidal oscillator that can be frequency modulated by a sawthoot/triangle waveform.
RAI Phonology Studio had one built by Bruel&Kiaer. It was originally used to demodulate radio signals.

C.FREQ controls the carrier frequency. The audible sinusoidal tone.
M.FREQ controls the modulator frequency.
Slope controls the shape of the modulator frequency. This runs from positive slope sawthoot, to triangle, to negative slope sawthoot.
Index controls the amount of the modulation.
Vol controls the carrier's signal output volume.

All in all this is a sine oscillator modulated by an LFO with FM technique.



The Square Oscillator produces square/rectangle waveforms. This kind of oscillator was not so common. Usually in the 50s, an average electronic studio had only one of these, most of the times it was a sine/square oscillator like the Heathkit AG10 used at WDR by Karheinz Stockhausen.

Range- 1 - 6 Frequency range like sine osc. 1
Duty Cycle - Simmetry of the waveform.
Vol - Output volume.



The Tone Burst Generator is (once again) another piece of equipment borrowed from radio laboratories. Its function in the electronic studio is comparable to a timed audio gate. When fed with a signal, the TBG will open and close accordingly to the two time parameters.

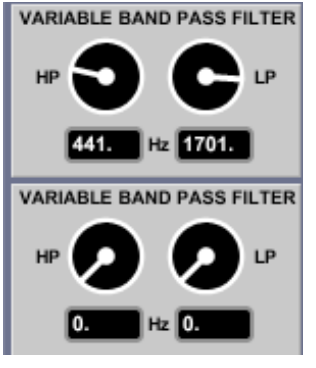
Closed- (2-4-8-16-32-64-128 Hz or whatever ms) sets the time the gate is closed
Open- (2-4-8-16-32-64-128 Hz) sets the time the gate is open

This version of the TBG will be soon updated.
The General Radio Company TBG 1398A had a second signal input that triggers the TBG. I'm working to make this TBG working almost the same way.



The Variable Band Pass Filters are made using HPF and LPF filters in serial connection. (with a 24dB/Oct slope) They're identical and the cutoff frequency (like almost every frequency setting) can be fine tuned and pushed up to 20000 Hz using the numerical display.

These two were not used in the 1950s, we could find them in 1968 at the RAI studio (Krohn-Hite 310A and 310B). Although including them is not entirely historically accurate, I thought it would have been nice to have them so to reproduce some parts of the Ricercare Elettronico by Roman Vlad.



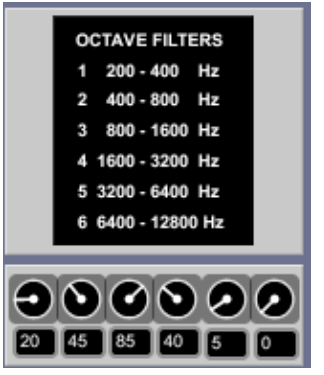
The Selective Band Pass Filter was another idea of Alfredo Lietti and it was built with the help of a frequency analyzer, high frequency (50KHz) oscillators, a filter and some kind of balanced ring modulator. Basically the result was a very narrow band pass filter (the original filter had a 2Hz bandwidth).

The Selective BPF was used mostly with the noise generators to obtain a "lively" sinewave. The noise in conjunction with this machine created a flute like tone, not as static (in amplitude) as the one produced by the sine oscillators. Luciano Berio used this filter (with voice samples) for his composition "Thema, omaggio a Joyce"[4]



[4] Evoluzione dei mezzi tecnici per la musica elettronica - Alfredo Lietti - 1959

The Octave Filters are 6 band pass filters in parallel, each one with its own amplitude control.
The cutoff frequencies have been copied from the Lietti's Octave Filters used at RAI in 1956[5].
The filters here have a 24dB/Oct slope. Please note I was not able to trace the original slope of these filters.

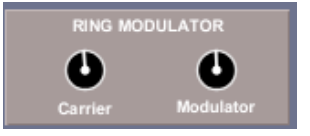


[5] Gli impianti tecnici dello studio di fonologia musicale di radio Milano - Alfredo Lietti

The Ring Modulator was a very common sound modifier at that time. This one has just two controls, one for the carrier signal's amplitude and one for the modulator signal's amplitude.

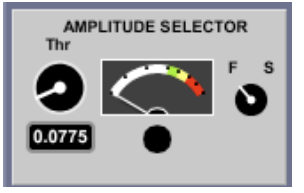
Ring modulation is basically frequency multiplication. The output is a complex timbre made of the sum and difference of the two frequencies used. If complex signals are used, sum and difference occours for every partial of both spectras.

Mind the daleks...



The Amplitude Selector is basically an audio gate.
As a signal goes over a certain threshold, the gate opens.
When it goes below, the gate closes (two different speeds are selectable Fast and Slow).

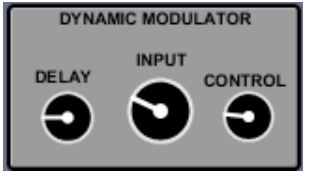
This machine was often used at RAI in conjunction with the Selective BPF and the Noise Generator to create random burst of filtered noise. Henry Pousseur used this device while creating his composition: “Scambi”[6].
You can set the Threshold and the close timing.



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[6] Evoluzione dei mezzi tecnici per la musica elettronica - Alfredo Lietti - 1959

Just like the device before, the Dynamic Modulator was designed by Alfredo Lietti for the RAI sudio. The scope of this device is to apply the envelope of a signal to another one. We call this envelope follower today.
INPUT: input volume of the signal to be modulated
DELAY: delay of the key signal (0 - 4 sec)
CONTROL: shape of release time



The Amplitude Modulator is basically a VCA modulated by an LFO. Any signal that goes here is modulated in its amplitude by a sinewave oscillator (0.5 - 20Hz).
The only control here is the frequency of the modulator signal.



The Frequency Shifter is also known as single sideband modulation. Traditionally this circuit was made using the heterodine effect through a complex scheme of modulations (multiplication) and phase modifications.
It shares many properties with the traditional ring modulator.

This frequency shifter ouputs the positive sidebands only. The amount of frequency shift is controlled by a dial and frequency display for fine tuning.

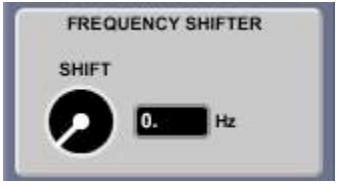
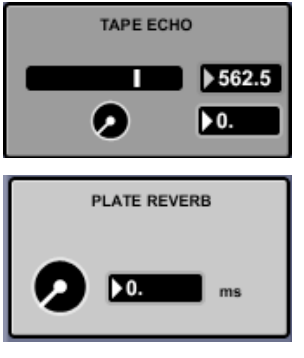


Plate Reverb and Tape Echo can be used from the mixer. Although no Tape Echo was used neither by WDR nor RAI, I've decided to include it as the echoplex dates back 1959. Normally tape echo in studios like RTF, WDR and RAI was made with tape recorders.

The plate reverb has a reverb time ranging from 0 to 4 seconds while the tape echo is from 0 to 670ms.
The feedback control range is from 0 to 0.99.
(0.99 = infinite repetitions).

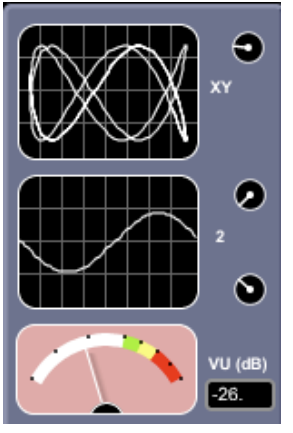


CHAPTER 6 SIGNAL MONITORING TOOLS

Signals in Berna can be visually monitored in three ways. At the top there is an Oscilloscope (OSCILLOSCOPE X and Y in the matrix) made to monitor the harmonic relationship between two or more sine oscillators. When there is an harmonic relationship (let's say between 100 and 200Hz) typical Lissajous figures will show up steady. The right dial controls the amplification.

The second Oscilloscope (OSCILLOSCOPE 2 in the matrix) is a one channel oscilloscope to monitor waveforms. The top dial controls the amplification, while the bottom dial controls the timing.

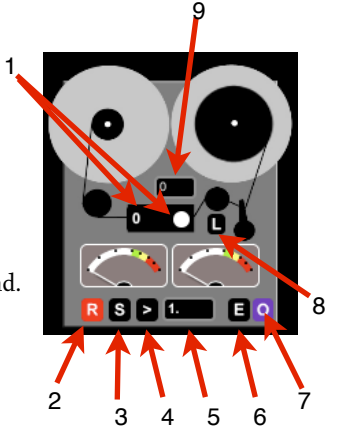
The third monitor is a VU meter, with a dB display.



CHAPTER 7 TAPE RECORDERS

Berna has four “tape” recorders.
All of them are identical.
1) Buffer time. When you want to record some sound you need to set the buffer (the space of your RAM) needed for the recording (default is 1 minute).
Select first how many seconds (or minutes) of time you want to record, then push the white dot.
2) R = record
3) S = stop
4) > = play
5) 0.5 to 1.5 = speed of the tape 1= normal
6) E = export the recording as an Aiff file
7) O = open an Aiff or Wav file
8) L = loop ON/OFF
(loop leght depends on the buffer you have choosed)
9) Seconds of play/record time elapsed - does not work if loop is activated and it is not affected by speed.

Tip
Record something on tape1 then send the master fader to tape 1 again. Turn loop on. Press play and rec together. Change speed or process sound.



CHAPTER 8 THE MIXER

The mixer is pretty simple.
There are 8 mono channels coming from the matrix. Each one has PAN, Reverb Send, Tape Echo Send.

In the center there is a red Master Fader of the 8 Channels. You can send the mix to one of the four tape recorders. On the right of the Master Fader there are the Reverb and Tape Echo returns channels.

Then 4 stereo channels coming from the tape recorders. Each one can be assigned to the master fader, or to the matrix.

If you need to play 2, 3 or 4 tape players at once in sync, you can use the SYNC PLAY FUNCTION.
Select the Tape Players you want to run in sync and then press “>” (Play) or “S” (Stop).



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by Paul Hegarty
Continuum 2008

Electronic and experimental music by Tom Holmes - Routledge, 2002

MIDI IMPLEMENTATION CHART

Oscillator 1 Frequency	CH1 CC1
Oscillator 1 Range	CH1 CC2
Oscillator 1 Volume	CH1 CC21
Oscillator 2 Frequency	CH1 CC3
Oscillator 2 Range	CH1 CC4
Oscillator 2 Volume	CH1 CC22
Oscillator 3 Frequency	CH1 CC5
Oscillator 3 Range	CH1 CC6
Oscillator 3 Volume	CH1 CC23
Oscillator 4 Frequency	CH1 CC7
Oscillator 4 Range	CH1 CC8
Oscillator 4 Volume	CH1 CC24
Oscillator 5 Frequency	CH1 CC9
Oscillator 5 Range	CH1 CC10
Oscillator 5 Volume	CH1 CC25
Oscillator 6 Frequency	CH1 CC11
Oscillator 6 Range	CH1 CC12
Oscillator 6 Volume	CH1 CC26
Oscillator 7 Frequency	CH1 CC13
Oscillator 7 Range	CH1 CC14

Oscillator 7 Volume	CH1 CC27
Oscillator 8 Frequency	CH1 CC15
Oscillator 8 Range	CH1 CC16
Oscillator 8 Volume	CH1 CC28
Oscillator 9 Frequency	CH1 CC17
Oscillator 9 Range	CH1 CC18
Oscillator 9 Volume	CH1 CC29
Noise Generator Volume	CH1 CC19
Noise Gen. White/Pink	CH1 CC20 (126/127)

Beat Osc Slope	CH1 CC30
Beat Osc Index	CH1 CC31
Beat Osc C.FFREQ	CH1 CC32
Beat Osc M.FFREQ	CH1 CC33
Beat Osc Volume	CH1 CC34
Tone Burst Close time	CH1 CC35
Tone Burst Open time	CH1 CC36
VarBPF1 HPF	CH1 CC37
VarBPF1 LPF	CH1 CC38
VarBPF2 HPF	CH1 CC39
VarBPF2 LPF	CH1 CC40

Selective BPF Frequency	CH1 CC106
Octave filters 1 vol	CH1 CC41
Octave filters 2 vol	CH1 CC42
Octave filters 3 vol	CH1 CC43
Octave filters 4 vol	CH1 CC44
Octave filters 5 vol	CH1 CC45
Octave filters 6 vol	CH1 CC46
Ring Mod Car Vol	CH1 CC47
Ring Mod Mod Vol	CH1 CC48
Amp Sel Threshold	CH1 CC49
Amp Sel rel. time	CH1 CC50 (126/127)
Dyn Mod Delay	CH1 CC51
Dyn Mod Input Vol	CH1 CC52
Dyn Mod Control	CH1 CC53
Frequency Shifter	CH1 CC54
Amplitude modulator	CH1 CC55
Square Osc Range	CH1 CC56
Square Osc Duty	CH1 CC57
Square Osc Volume	CH1 CC58
Square Osc Frequency	CH1 CC59

Plate reverb time	CH1 CC60
Tape Echo Time	CH1 CC61
Tape Echo FBK	CH1 CC62
Mixer CH1 Fader	CH1 CC63
Mixer CH1 Pan	CH1 CC78
Mixer CH1 Rev Send	CH1 CC86
Mixer CH1 TE Send	CH1 CC94
Mixer CH2 Fader	CH1 CC64
Mixer CH2 Pan	CH1 CC79
Mixer CH2 Rev Send	CH1 CC87
Mixer CH2 TE Send	CH1 CC95
Mixer CH3 Fader	CH1 CC65
Mixer CH3 Pan	CH1 CC80
Mixer CH3 Rev Send	CH1 CC88
Mixer CH3 TE Send	CH1 CC96
Mixer CH4 Fader	CH1 CC66
Mixer CH4 Pan	CH1 CC81
Mixer CH4 Rev Send	CH1 CC89
Mixer CH4 TE Send	CH1 CC97
Mixer CH5 Fader	CH1 CC67
Mixer CH5 Pan	CH1 CC82
Mixer CH5 Rev Send	CH1 CC90

Mixer CH5 TE Send	CH1 CC98
Mixer CH6 Fader	CH1 CC68
Mixer CH6 Pan	CH1 CC83
Mixer CH5 Rev Send	CH1 CC91
Mixer CH5 TE Send	CH1 CC99
Mixer CH7 Fader	CH1 CC69
Mixer CH7 Pan	CH1 CC84
Mixer CH5 Rev Send	CH1 CC92
Mixer CH5 TE Send	CH1 CC100
Mixer CH8 Fader	CH1 CC70
Mixer CH8 Pan	CH1 CC85
Mixer CH5 Rev Send	CH1 CC93

Mixer CH5 TE Send	CH1 CC101
Mixer Master Fader	CH1 CC71
Mixer Reverb Return	CH1 CC72
Mixer Tape Echo Return	CH1 CC73
Mixer Tape 1 Return	CH1 CC74
Mixer Tape 2 Return	CH1 CC75
Mixer Tape 3 Return	CH1 CC76
Mixer Tape 4 Return	CH1 CC77
Tape 1 speed	CH1 CC102
Tape 2 speed	CH1 CC103
Tape 3 speed	CH1 CC104
Tape 4 speed	CH1 CC105

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